

4.03 Matrices				
4.03a	The language of matrices	a) Understand the language of matrices. <i>Understand the meaning of “conformable”, “equal”, “square”, “rectangular”, “m by n”, “determinant”, “zero” and “null”, “transpose” and “identity” when applied to matrices.</i> <i>Learners should be familiar with real matrices and complex matrices.</i>		C2
4.03b	Matrix addition and multiplication	b) Be able to add, subtract and multiply conformable matrices; multiply a matrix by a scalar. <i>Learners may perform any operations involving entirely numerical matrices by calculator.</i> <i>Includes raising square matrices to positive integer powers.</i> <i>Learners should understand the effects on a matrix of adding the zero matrix to it, multiplying it by the zero matrix and multiplying it by the identity matrix.</i>		C1
4.03c		c) Understand that matrix multiplication is associative but not commutative. <i>Understand the terms “associative” and “commutative”.</i>		

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...	DfE Ref.
4.03d	Linear transformations	d) Be able to find and use matrices to represent linear transformations in 2-D. <i>Includes:</i> <i>reflection in either coordinate axis and in the lines $y = \pm x$</i> <i>rotation about the origin (defined by the angle of rotation θ, where the direction of positive rotation is taken to be anticlockwise)</i> <i>enlargement centre the origin (defined by the the scale factor)</i> <i>stretch parallel to either coordinate axis (defined by the invariant axis and scale factor)</i> <i>shear parallel to either coordinate axis (defined by the invariant axis and the image of a transformed point).</i> <i>Includes the terms “object” and “image”.</i>		C3
4.03e		e) Be able to find and use matrices to represent successive transformations. <i>Includes understanding and being able to use the result that the matrix product $\mathbf{AB}$ represents the transformation that results from the transformation represented by $\mathbf{B}$ followed by the transformation represented by $\mathbf{A}$.</i>		

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4.03f		f) Be able to use matrices to represent single linear transformations in 3-D. <i>3-D transformations will be confined to reflection in one of the planes $x = 0$, $y = 0$, $z = 0$ or rotation about one of the coordinate axes. The direction of positive rotation is taken to be anticlockwise when looking towards the origin from the positive side of the axis of rotation.</i> <i>Includes the terms “plane of reflection” and “axis of rotation”.</i> <i>In stage 1 knowledge of 3-D vectors is assumed: see H240 section 1.10b.</i>		C3
4.03g	Invariance	g) Be able to find invariant points and lines for a linear transformation. <i>Includes the distinction between invariant lines and lines of invariant points.</i> <i>[The 3-D transformations in section 4.03f are excluded.]</i>		C4
4.03h 4.03i	Determinants	h) Be able to find the determinant of a 2×2 matrix with and without a calculator. <i>Use and understand the notation $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ or $\mathbf{M}$ or $\det \mathbf{M}$.</i> i) Know that the determinant of a 2×2 matrix is the area scale factor of the transformation defined by that matrix, including the effect on the orientation of the image. <i>Learners should know that a transformation preserves the orientation of the object if the determinant of the matrix which represents it is positive and that the transformation reverses orientation if the determinant</i>		C5

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4.03j		j) Be able to calculate the determinant of a 3×3 matrix with and without a calculator.		C5
4.03k		k) Know that the determinant of a 3×3 matrix is the volume scale factor of the transformation defined by that matrix, including the effect on the preservation of the orientation of the image. <i>Learners should know that the sign of the determinant determines whether or not the corresponding transformation preserves orientation, but do not need to understand the geometric interpretation of this in 3-D.</i>		
4.03l		l) Understand and be able to use singular and non-singular matrices. <i>Includes understanding the significance of a zero determinant.</i>		
4.03m		m) Know and be able to use the result that $\det(\mathbf{AB}) = \det(\mathbf{A}) \times \det(\mathbf{B})$.		
4.03n	Inverses	n) Be able to find and use the inverse of a non-singular 2×2 matrix with and without a calculator.		C6
4.03o		o) Be able to find and use the inverse of a non-singular 3×3 matrix with and without a calculator.		
4.03p		p) Understand and be able to use simple properties of inverse matrices. <i>e.g. The result that $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$.</i>		
4.03q		q) Understand and be able to use the connection between inverse matrices and inverse transformations.		

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4.03r 4.03s	Solution of simultaneous equations	r) Be able to solve two or three linear simultaneous equations in two or three variables by the use of an inverse matrix, where a unique solution exists.	s) Be able to determine, for two or three linear simultaneous equations where no unique solution exists, whether the equations have an infinite set of solutions (the equations are consistent) or no solutions (the equations are inconsistent). <i>[Finding the solution set in the infinite case is excluded.]</i>	C7
4.03t	Intersection of planes		t) Be able to interpret the solution or failure of solution of three simultaneous linear equations in terms of the geometrical arrangement of three planes. <i>Learners should know and be able to identify the different ways in which two or three distinct planes can intersect in 3-D space, including cases where two or three of the planes are parallel.</i> <i>Learners should understand and be able to apply the geometric significance of the singularity of a matrix in relation to the solution(s) or non-existence of them.</i> <i>[Finding the line of intersection of two or more planes is excluded.]</i>	C8